

SHAUNAK RANE

AI/ML Engineer | Backend & LLM Systems | Python | FastAPI

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PROFESSIONAL SUMMARY

AI and software engineering intern with proven experience delivering end-to-end solutions across full Software Development Lifecycle (SDLC): from requirements analysis and system design to implementation, testing, and deployment. Strong analytical and problem-solving skills applied to real client workloads, autonomous multi-agent systems, LLM API pipelines, and production-grade machine learning services. Collaborative team contributor with hands-on background in Python, REST APIs, SQL databases, Docker, and automated test engineering.

TECHNICAL SKILLS

AI & LLM Systems: Agentic AI, Multi-Agent Workflows, RAG, Prompt Engineering, Structured Outputs (JSON Schema), Function & Tool Calling, Vector Embeddings, LLM APIs (Gemini, OpenAI)

Machine Learning & Data: PyTorch, PyTorch Geometric, GraphSAGE, Scikit-learn, Statsmodels, NumPy, Pandas, Exploratory Data Analysis, Time-Series Modeling, Anomaly Detection

Backend & Languages: Python, C, SQL, TypeScript/JavaScript, FastAPI, Flask, RESTful APIs, WebSockets, OOP, SDLC, System Design

Databases & DevOps: PostgreSQL, Supabase (pgvector), MySQL, Docker, Git, GitHub Actions, Linux, Postman, pytest, CI/CD

Engineering Practices: Full SDLC, Test-Driven Development (TDD), Causal Evaluation, API Observability, Agile Collaboration, Microservices, Documentation

WORK EXPERIENCE

Univitt Technologies - *AI Engineering Intern*

May 2026 - Aug 2026 | Thane, India (Remote)

- Architected end-to-end predictive analytics pipeline for centrifugal compressor Condition-Based Monitoring (CBM) across 3 industrial facilities and 4,000+ daily historian records.
- Reproduced 3,879 historical Degradation Factor Index (DFI) rows to $<1e-10$ absolute error; engineered causal maintenance-event logic with 17/17 regression checks passing.
- Built and containerized REST inference microservices in Docker with Flask, automating anomaly detection alerts with structured API responses.
- Audited 400+ physical and mechanical process variables, evaluated ML vs. statistical forecasting baselines, and instituted automated promotion gates for model deployment.

Univitt Technologies - *Software Engineering Intern*

Apr 2025 - Jul 2025 | Thane, India (Remote)

- Engineered high-throughput RESTful backend APIs and database schemas with FastAPI and PostgreSQL for an institutional food operations and cost-management platform (Sodexo).
- Developed end-to-end automated workflows for menu generation, Bill-of-Materials (BOM) cost calculations, live inventory tracking, and operator dashboards.
- Integrated a Random Forest predictive attendance model into operational shift scheduling, achieving 88% forecast accuracy across workforce shifts.
- Implemented comprehensive test suites with pytest, executed schema migrations, and collaborated on code reviews using Git and GitHub workflows.

PROJECTS

Project Aegis - *Autonomous Multi-Agent Triage System*

Python, FastAPI, Agentic AI, RAG, Supabase (pgvector), Docker

Live Demo: agenticai914.netlify.app | Source Code: github.com/Sh Shaunakrane914/Misinformation

- Built a multi-agent incident-triage and response simulation with hybrid RAG retrieval and tool-calling agent swarms, enforcing strict JSON Schema validation for deterministic structured outputs.
- Engineered tool-calling agent workflows to autonomously query incident records, evaluate severity levels, and route prioritized response actions across the multi-agent pipeline.
- Implemented semantic vector persistence via Supabase (pgvector), API-key isolation, and structured audit trails for agent-to-agent message passing and event reporting.

Gridium Protocol - *Multi-Agent Microgrid Optimization*

Python, PyTorch, FastAPI, DDPG, WebSockets

Live Demo: live-ai-1-7tcy.vercel.app | Source Code: github.com/Sh Shaunakrane914/Live-Ai-1

- Developed a multi-agent reinforcement learning engine utilizing Deep Deterministic Policy Gradients (DDPG) to dynamically optimize power routing across a 15-node simulated microgrid.
- Coordinated decentralized agent policies managing solar generation, battery storage, and dynamic loads, achieving an 18.4% reduction in peak-hour grid strain in simulation.
- Built real-time telemetry streaming pipelines using FastAPI and WebSockets, feeding continuous operational metrics to interactive monitoring dashboards.

TopoFlow - *GNN Fluid Dynamics Simulation*

Python, PyTorch Geometric, GraphSAGE, FastAPI

Source Code: github.com/Sh Shaunakrane914/Flow

- Benchmarked GraphSAGE against Kozeny-Carman physics baselines across 1,231 pore-network samples from 5 geological formations, using pore-size heterogeneity to dictate the modeling regime.
- Reduced MSE loss by 46.2% on Savonnières and 28.4% on Estailades formations while maintaining rigorous physical consistency benchmarks across baseline solvers.
- Engineered pore-network extraction pipelines, GNN model training workflows, and FastAPI inference microservices with PyTorch Geometric.

EDUCATION

Universal AI University - *B.Tech in Artificial Intelligence & Machine Learning*

Expected May 2028 | Karjat, Maharashtra

- CGPA: 8.47** | **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Deep Learning, DBMS, Operating Systems, Linear Algebra & Probability.